(*i*) Estimation

Students should be taught to:

estimate population means from samples;

estimate population proportions from samples with applications in opinion polls and elsewhere;

estimate population size based on the Petersen capture/recapture method;

The appropriateness of the assumptions in practice.

understand the effect of sample size on estimates and the variability of estimates, **with a simple quantitative appreciation of appropriate sample size**.

3 Reasoning, interpreting and discussing results

Content	Notes
Students should be taught in the context of real data to:	
apply statistical reasoning, explain and justify inferences, deductions, arguments, solutions **and decisions;**	Cases clearly restricted to the content of the specification at the appropriate level
explore connections, look for and examine relationships between variables **including fitting the equation to a line of best fit or trend line**;	Eg height and weight, age and depreciation of a car, GNP and mortality in infants. **Interpretations of gradient and intercept will be expected**.
consider the limitations of any assumptions;	Simple cases only, eg honest replies to questionnaires, equally likely outcomes in probabilities, representativeness of sample of population, reliability of secondary data.
formally identify outliers using quartiles;	**Dealing with outliers will be expected.**
relate summarised data to any initial questions or observations;	The relevance of measures of central tendency.
interpret all forms of statistical tables, diagrams and graphs;	To include real published tables and graphs.
compare distributions of data and make comparisons using measures of central tendency and measures of dispersion, percentiles, **deciles, inter-percentile range, mean deviation, variance and standard deviation**;	The shapes of distributions and graphs may be used. Formula for variance and standard deviation to be given.
check results for reasonableness and modify their approaches if necessary;	Eg the mean must lie between the maximum and minimum, 'the average bicycle speed was 130 km per hour' is not reasonable.
interpret correlation as a measure of the strength of the association between two variables, **including Spearman's rank correlation coefficient for ranked data;**	The use of words such as weak or strong will be expected; **the closer to ± 1 the better the correlation for a given sample size. Beware the use of correlation in small samples.**
make predictions;	**The use of a trend line by eye, drawing or formula will be expected.**
compare or choose by eye between a line of best fit and a model based on $y = ax^n + b$ for $n = 2, 1$ or ½, $y = ax^2 + bx$ or $y = ka^x$.	**Based on an informal awareness of the spread of points around a proposed model.**

4 Probability

Content	Notes
Students should be taught to:	
understand the meaning of the words event and outcome;	Tossing a coin is an event with outcomes landing Heads or Tails.
understand words such as impossible, certain, highly likely, likely, unlikely, possible, evens and present these on a likelihood scale;	Interpretation of real-life situations will be expected, eg ‘the probability that the horse will win the next race is 0.3’; ‘the probability that I will get a grade C or better in my Statistics in 3/4.’
put outcomes in order in terms of probability;	Use of $\leq$ will be expected.
put probabilities in order on a probability scale;	Labelling of the scale will be expected.
understand the terms ‘random’ and ‘equally likely’;	
understand and use measures of probability from a theoretical perspective and from a limiting frequency or experimental approach;	Formal definition and notation of a limit will not be required whilst terminology such as ‘as the number of trials increases’ will be required. **Understand that increasing sample size generally leads to better estimates of probability and population parameters.**
the relationship between ‘odds’ and probability;	**The relationship between $\Sigma p = 1$ and ‘odds’ may be tested.**
understand that in some cases the measure of probability based on limiting frequency is the only viable measure;	The probability of a sports team winning can only be measured from a limiting frequency perspective. For example, medical statistics for assessment of health risks.
compare expected frequencies and actual frequencies;	
use simple cases of the binomial and discrete uniform distribution;	**The expansion of $(p + q)^2$ will be expected. In all other cases the expansion of $(p + q)^n$ will be given.**
use simulation to estimate more complex probabilities;	
use probability to assess risk;	Examples may be taken from insurance.
produce, understand and use a sample space;	Listing all outcomes of single events and two successive events, in a systematic way.
understand and use Venn diagrams and cartesian grids;	

Content	Notes
understand the terms mutually exclusive and exhaustive and to understand the addition law P(A or B) = P(A) + P(B);	P(A or B) = P(A) + P(B); 'Mutually exclusive' means that the occurrence of one outcome prevents another, $\Sigma = 1$ when summed over all mutually exclusive outcomes.
know, for mutually exclusive outcomes, that the sum of the probabilities is one and in particular the probability of something not happening is one minus the probability of it happening;	If P(A) = p then P(not A) = $1 - p$.
form and use tree diagrams and probability tree diagrams for independent events **and conditional cases**;	Listing all possible joint or compound outcomes **with and without replacement for up to three outcomes and three sets of branches.**
understand, use and apply the addition, **general addition** and multiplication laws for independent events **and conditional events and outcomes;**	To correctly apply P(A and B) = P(A) × P(B), P(A or B) = P(A) + P(B) **P($A \cup B$) = P(A) × P(B) – P($A \cap B$),** **P(A or B) = P($B \mid A$) × P(A).**
the shape and simple properties of the normal distribution.	**The distribution is symmetrical with mean, mode and median equal; approximately 95% of values are within ± 2 standard deviations of the mean; virtually all values are within ± 3 standard deviations of the mean. Use of the normal distribution to model some populations.** **Use of Normal distribution tables will not be required.**

Edexcel GCSE Statistics

Formulae Sheet

Foundation Tier

Mean of a frequency distribution = $\dfrac{\sum fx}{\sum f}$.

Mean of a grouped frequency distribution = $\dfrac{\sum fx}{\sum f}$, where x is the mid-interval value.

Edexcel GCSE Statistics

Formulae Sheet

Higher Tier

Mean of a frequency distribution $= \dfrac{\sum fx}{\sum f}$.

Mean of a grouped frequency distribution $= \dfrac{\sum fx}{\sum f}$, where x is the mid-interval value.

Variance $= \dfrac{\sum (x-\bar{x})^2}{n}$

Standard deviation (set of numbers) $\sqrt{\left[\dfrac{\sum x^2}{n} - \left(\dfrac{\sum x}{n}\right)^2\right]}$

or $\sqrt{\left[\dfrac{\sum (x-\bar{x})^2}{n}\right]}$

where $\bar{x}$ is the mean set of values.

Standard deviation (discrete frequency distribution) $\sqrt{\left[\dfrac{\sum fx^2}{\sum f} - \left(\dfrac{\sum fx}{\sum f}\right)^2\right]}$

or $\sqrt{\left[\dfrac{\sum f(x-\bar{x})^2}{\sum f}\right]}$

Spearman's rank correlation coefficient $1 - \dfrac{6\sum d^2}{n(n^2-1)}$